MOHAMED ABBAS MOYNUDEEN

PRINCIPAL ENGINEER | TECHNICAL LEADER

CONTACT

+91 8754264561

mohamedabbas.bsc@gmail.com

https://www.linkedin.com/in/abbas-md/

228, Chinna Samy Nagar, Tennur,
Trichy, Tamil Nadu - 620017

SKILLS

Python, Terraform, AWS, Data Processing, SQL, Deep Learning, Machine Learning, LLM Models, Leadership, Team Management, Product Management

EDUCATION

Master of Computer Application

Bharathidasan University

2018-2020

Bachelor of computer Science

Jamal Mohamed College

2014 - 2017

REFERENCES

Darren Breen - Senior Engineer Manager

+44 7848885189

Darren.Breen@woodmac.com

David Kwa - Director Woodmac

david.kwa@woodmac.com

PROFILE

Principal Engineer with 10+ years of experience in technical leadership, product architecture, and AI-driven automation, currently managing 20+ engineers across 4 teams and 2 managers. Adept at driving large-scale enterprise initiatives in data engineering, machine learning, and generative AI, with a proven ability to design and deliver scalable, high-impact platforms.

Recognized as a company-wide AI Specialist, I have spearheaded flagship programs such as the AI Asset Tracker (enterprise-scale LLM-powered knowledge management platform), ETP pipeline infrastructure modernization, and neuro-symbolic model retraining initiatives.

I work closely with Directors, Vice Presidents, and global clients, leading cross-functional collaboration, roadmap planning, and strategic execution. My leadership approach combines deep technical expertise (AI/ML, cloud-native architectures, LLMs, data pipelines) with a focus on people development and business value delivery.

Key strengths include:

- Strategic Leadership – managing multi-team organizations, influencing technical roadmaps, aligning engineering with business goals.
- AI & Innovation – architecting and scaling LLM-driven platforms, retraining pipelines, and generative AI products.
- Operational Excellence – building resilient ETL pipelines, quality validation frameworks, and observability systems.
- Mentorship & Growth – coaching engineers and managers, improving productivity, and enabling upskilling across teams.

Key Technical Skills

- Programming & Development: Python, Node.js, JavaScript, SQL, Flask REST APIs, Pandas, NumPy, GStreamer, WebRTC
- AI / ML / Deep Learning: TensorFlow, Keras, OpenCV, PyTorch, tesseract, dataset engineering, neuro-symbolic retraining, LLM fine-tuning (GPT, Claude, LLaMA)
- Generative AI & LLMs: Prompt engineering, RAG pipelines, enterprise LLM integration, model deployment, knowledge indexing (Titan, OpenSearch)
- Cloud & Infrastructure: AWS (Lambda, Step Functions, ECS, S3, DynamoDB, CloudWatch, Kendra, Bedrock), Terraform, containerized pipelines, observability & monitoring
- Data Engineering & Pipelines: Large-scale ETL design, scraper pipelines, QC/validation frameworks, log analysis platforms, anomaly detection
- Product & Architecture Design: High-level system design, AI-first architecture strategy, enterprise automation platforms, scaling solutions across business units

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WORK EXPERIENCE

Principle Engineer

Wood Mackenzie 2024 - Present

Product - AI Asset Tracker

Role: Principal Engineer & Technical Lead | Team Size: 5 engineers (core team) + cross-team collaboration (data, infra, product)

- Designed and led development of an AI-powered enterprise knowledge management and asset tracking platform, similar to Perplexity AI but customized for internal enterprise use with compliance and governance layers.

Architecture & Technologies:

- AWS: ECS, Lambda, Step Functions, DynamoDB, S3, CloudWatch, Bedrock, Kendra.
- Vector Search & Indexing: OpenSearch with Titan embeddings.
- LLMs: Integrated Claude 3.5, GPT, LLaMA, and fine-tuned open-source models for prediction and Q&A.
- ETL Layer: Automated ingestion pipelines with scrapers, validators, and knowledge base lookups.
- Observability: Log analysis + anomaly detection integrated with CloudWatch/Splunk.

Key Features Delivered:

- Built semantic vector search pipelines enabling natural language queries over large knowledge bases.
- Designed dual-lookup system → primary knowledge base search + Google fallback integration for missing queries.
- Implemented neuro-symbolic retraining pipeline for continuous LLM fine-tuning on company-specific data.
- Created prediction & reporting flows powered by Claude 3.5 for insights and decision-making.
- Architected hyperlinked reporting engine with source mapping and unit normalization.

Leadership & Delivery:

- Led 5 core engineers while collaborating with 4 teams (~30 people) across infra, data engineering, and product.
- Mentored engineers on vector databases, LLM integration, and scalable cloud-native design.
- Coordinated with Directors, VPs, and client stakeholders for feature planning, roadmap alignment, and delivery execution.
- Business Impact:
- Reduced manual knowledge lookup effort by ~70%, speeding up research and decision-making.
- Increased cross-business-unit data reusability, cutting duplication costs.
- Established company branding as an AI-first automation leader, positioning the platform as a flagship AI project.

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Project - WM – QC Tool & Validation Framework (2022 – 2024)

Role: Principal Engineer & Initiative Owner | Team Size: 6 engineers

- Developed a data quality validation framework to ensure accuracy and completeness of scraped, processed, and analyzed datasets. The tool automated a previously manual, error-prone validation process, improving productivity and trust in downstream analytics.

- **Architecture & Technologies:**
  - Core Stack: Python scripts, AWS Glue, Glue Scripts, Athena, Lambda, Step Functions.
  - **Validation Logic:**
    - Missing fields (dates, values, records).
    - Pattern mismatches and schema deviations.
    - Record count consistency across pipeline stages.
    - Statistical anomaly detection.
  - **AI Integration:** ~20% automation with Bedrock-powered anomaly detection models; majority of checks implemented programmatically for efficiency.
  - **Reporting & Visibility:**
    - Generated detailed validation reports with missing percentage & anomaly breakdowns.
    - Automated report distribution to Teams channels via Power Automate flows, ensuring real-time visibility for stakeholders.

- **Key Features Delivered:**
  - End-to-end automated QC layer integrated into scraping, processing, and analysis pipelines.
  - Flexible rule engine (programmatic + AI) allowing teams to extend validation logic easily.
  - Team-wide visibility through Power Automate-driven notifications and reports.

- **Leadership & Delivery:**
  - Led a 6-member cross-functional team (data engineering + QA) to design, implement, and operationalize QC tool.
  - Introduced best practices for validation-first development across scraping and pipeline teams.
  - Mentored engineers on Glue scripting, Athena queries, and automation workflows.

- **Business Impact:**
  - Reduced manual validation effort by 70%, freeing engineers to focus on development.
  - Improved data reliability and trustworthiness across scrapers and ETL pipelines.
  - Enhanced team productivity with automated visibility flows → fewer escalations, faster release approvals.

Designed and implemented a data quality validation framework that significantly improved delivery reliability. Reduced production defects by ~60%, accelerated issue resolution time by 70%, and boosted pipeline success rate from 82% to 95%. Stakeholder trust increased by 40% (survey-based), while release approval timelines dropped from weeks to days, enabling faster, more confident deployments.

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Project : LDP – Lens Data Platform (2022 – 2024)

Role: Principal Engineer
(transitioned from Contractor → Wood Mackenzie FTE after project success)
Team Size: 8 engineers

Architected and led the development of Lens Data Platform (LDP), a large-scale data ingestion and processing system designed to scrape, process, and validate energy-sector datasets from multiple global sources. The success of this project directly resulted in my promotion from a contractor role at UnifyCX (formerly GlowTouch) to a Principal Engineer position at Wood Mackenzie.

- **Architecture & Technologies:**
  - Scraping & Processing: Python-based scraper framework, deployed on AWS ECS, Stepfunction, Lambda, Glue.
  - Pipeline Orchestration: Step Functions + Lambda for fault-tolerant, multi-stage orchestration.
  - Data Storage & Querying: S3 (raw/processed zones), Glue ETL scripts, Athena queries.
  - Data Quality Validation: Full integration with the QC Tool, ensuring missing values, anomalies, and schema mismatches were automatically flagged and resolved.
  - Automation & Reusability: Designed Terraform “Shared Deployment Modules” reusable modules for deploying scrapers and ETL pipelines, accelerating onboarding of new datasets.

- **Key Features Delivered:**
  - Developed an advanced scraping system capable of handling geo-restrictions, proxy limitations, and API compliance requirements.
  - Built auto-healing pipelines → failed scrapers retried automatically; missing data triggered backfill pipelines.
  - Established a pluggable architecture → new scrapers/ETL pipelines onboarded 60% faster.
  - Provided real-time QC-driven validation reports shared via Teams (Power Automate flows) for business visibility.

- **Leadership & Delivery:**
  - Led an 8-member data engineering team, guiding both technical delivery & strategic roadmap alignment.
  - Acted as the bridge between clients, business teams, and engineering, ensuring delivery aligned with business-critical energy datasets.
  - Mentored engineers on pipeline observability, automation-first principles, and validation frameworks.
- **Business & Career Impact:**
  - Automated ingestion of 15+ mission-critical energy forecast datasets used across business lines.
  - Improved data accuracy, completeness, and reliability → strengthened client trust and accelerated decision-making.
  - Accelerated delivery cycles by 60% through Terraform automation and modularized pipeline design.
  - This project’s success was recognized as a key transformation milestone, leading to my transition from Contractor at GlowTouch/UnifyCX → Principal Engineer at Wood Mackenzie.

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Innovation & POCs (Proof of Concepts)

In addition to major projects, I proactively initiated and delivered multiple POCs to improve pipeline stability, developer productivity, and organizational efficiency. These were often implemented outside of regular project work, showcasing my commitment to innovation and continuous improvement.

Log Analysis & Observability System

- Designed and implemented a log aggregation and monitoring framework for ETL pipelines and scrapers.
- Built dashboards in Splunk + CloudWatch to provide real-time health monitoring.
- Developed an AI-driven error log analyzer → automatically detects failure patterns, links to codebase, and suggests corrective actions.
- Integrated with alerting system → stakeholders receive error context + AI-generated solutions.

AI-Assisted Developer Productivity (Copilot + MCP Server)

- Introduced GitHub Copilot adoption across engineering teams to accelerate development.
- Identified LLM hallucination risks in Copilot outputs; created instruction templates to reduce inaccuracies.
- Deployed an MCP server integration with VS Code for company-wide developers, binding curated instruction templates to enforce coding consistency.
- Improved developer efficiency while reducing rework caused by hallucinated suggestions.

Pipeline & Product Stability Enhancements

- Designed multiple automated fallback and retry mechanisms for scrapers and ETL pipelines.
- Introduced modular Terraform templates ("Shared Deployment Modules") for standardized, faster pipeline deployments.
- Improved overall system reliability and release stability, reducing pipeline failure rates and manual interventions.

Deep Learning Engineer

Solutionchamps Technologies | 2019 – 2021

- Served as core member of the 14-member **DVR AI BOX** development team, responsible for all AI components and backend services.

- **Recording Engine:**
  - Designed and implemented a high-quality video recording engine with support for WebRTC streaming, automatic clip deletion, URL probing, and timeline indexing.
  - Optimized encoding/decoding using NVDEC and NVENC hardware accelerators on Nvidia Jetson Nano/Xavier platforms.
  - Built storage solution with SQLite + Amazon S3; implemented tiered storage with Glacier backups.

- **Analytics Engine:**
  - Built a real-time analytics engine using GStreamer + NVDEC, performing continuous background subtraction on live streams.
  - Applied clustering and bounding box logic to detect movement, capture pre/post event images, and export metadata.
  - Trained and deployed a Tiny YOLO model optimized for Jetson boards, enabling lightweight, high-performance object detection.
  - Automated upload of detection results (images + clips) to Amazon S3, powering security alerting workflows.
- **Middleware & APIs:**
  - Integrated backend services with Node-RED middleware for API handling and orchestration.
  - Enabled UI-driven API interactions, including real-time playback, device management (Onvif), and customer alerting via email.
- **Tech Stack:** Nvidia Jetson Nano/Xavier, YOLO Darknet, GStreamer, FFMPEG, Node-RED, MongoDB, SQLite, Amazon S3/Glacier, Angular7, ASP.NET.
- **Impact**: Delivered a production-ready AI-powered DVR platform from the ground up, combining video analytics, real-time streaming, and automated storage — recognized as a key backend contributor for the company's flagship product.

Deep Learning Engineer

Solutionchamps Technologies | 2019 – 2021

Receipt Parsing with OCR (Team Lead, 6 Members)

- Led development of a receipt parsing system using OCR + object detection to extract structured fields (date, tax, total, line items) from mobile-captured receipts.
- Built Flask REST API pipeline for image ingestion → detection → parsing.
- Collected and annotated a custom dataset of 10,000 receipts; trained YOLOv4 model for field localization.
- Applied text classification & transfer learning techniques to improve recognition accuracy.
- Leadership: Acted as Team Leader & Code Integrator, coordinated sprint planning under agile methodology.
- Impact: Successfully deployed to production; personally resolved major deployment blockers, earning recognition for fast turnaround and coordination.

Recommendation Engine for E-Commerce (Team Lead, 6 Members)

- Led development of a personalized recommendation engine for an e-commerce platform.
- Designed and implemented recommendation logic using Apriori Algorithm in TensorFlow (Python).
- Combined Deep Learning (Keras, TensorFlow) with classical ML approaches to improve recommendation accuracy.
- Leadership: Acted as core development team leader & integrator, ensuring code quality and on-time delivery.
- Applied RPA techniques and agile practices to streamline development and deployment.
- Impact: Delivered a production-ready recommendation engine; resolved critical deployment issues quickly, earning recognition for team coordination and delivery ownership.

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Full Stack Developer

SurfaceInsight Technologies | 2018 – 2019

- Served as Team Leader & Code Integrator for a 6-member core development team, driving the design and development of RPA (Robotic Process Automation) products.
- Developed object detection features and automation workflows using Core Java, Python, Node.js, C++, TensorFlow, Keras, and OpenCV.
- Built and deployed REST APIs (Python/Java), integrated with Electron desktop cross-platform apps, and implemented frontend features using JavaScript, TypeScript, HTML, CSS.
- Designed data processing modules with SQL and JSON, enabling seamless integration between RPA automation components.
- Applied cloud services (AWS) for deployment and scaling, ensuring reliability and performance.
- Gained strong expertise in RPA methodologies and applied agile techniques for sprint-based delivery.
- Actively contributed to production releases, resolving a significant number of deployment issues under tight deadlines.
- Impact: Recognized for effective team coordination and delivery ownership, while significantly contributing to the company's flagship RPA product line.

Application Developer
Tapceipt IT Solution | 2017 – 2018

- Worked as Product Developer & Code Integrator in a 4-member core team building digital e-receipt solutions.
- Developed cross-platform applications using AngularJS, IONIC, JavaScript, TypeScript, and REST APIs.
- Contributed to backend services with Java + AWS integration, enabling secure cloud-based receipt management.
- Designed and implemented data storage & processing modules using SQL + JSON for structured receipt handling.
- Production Delivery: Played a key role in release cycles, resolving critical deployment issues in record time.
- Recognition: Earned accolades for effective team coordination and problem-solving under tight deadlines.

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TO WHOM IT MAY CONCERN,

I am Mohamed Abbas Moynudeen, a Principal Engineer and Technical Leader with 10+ years of experience in AI/ML, deep learning, RPA, and cloud-native data engineering. I currently manage 30+ engineers across 4 teams and 2 managers, working closely with Directors, VPs, and clients to align technical execution with business goals.

Key achievements include:

- Architecting the AI Asset Tracker (LLM-powered enterprise knowledge platform) and the Lens Data Platform (LDP) for large-scale energy data ingestion, both flagship AI initiatives at Wood Mackenzie.
- Building the QC Validation Tool, reducing production defects by 35% and improving trust in data pipelines.
- Delivering deep learning solutions at Solutionchamps (DVR AI Box, OCR-based receipt parsing, e-commerce recommendation engine).
- Leading RPA and full-stack development projects at SurfaceInsight, gaining strong expertise in automation and agile delivery.
- Driving innovation POCs such as AI-powered log analysis, Copilot/MCP integration for developer productivity, and Terraform automation modules.

With expertise in LLMs, deep learning, cloud pipelines, and automation, combined with proven leadership and people management skills.

Sincerely,
Mohamed Abbas Moynudeen